

GROUP D (EVE)



MAKERERE UNIVERSITY

GROUP D (EVENING) UNIT TESTING LAB ASSIGNMENT		
Name	Registration Number	Student Number
NAKALEMA JULIAN	23/13388/EVE	2300713388
KAYIWA RAHIM	23/09499/EVE	2300709499
NSUBUGA IBRAHIM	23/24436/EVE	2300724436
NAMEMBWA SHERRY	23/14586/EVE	2300714586
OGENRWOTH JIM FRANK	23/U/16562/PS	2300716562

Meeting Planner System. Test Plan & Results

1.Scope and Objectives

This test plan covers JUnit unit testing of the Meeting Planner system. The system manages calendar bookings for rooms and employees, preventing double-booking and tracking availability.

1.1. Features Under Test

- Booking a meeting (addMeeting, valid, conflicting, boundary dates)
- Booking vacation/holiday time (full-day meeting blocks)
- Checking availability for a room and a person (isBusy)
- Printing the agenda for a room and a person (printAgenda)
- Removing and retrieving meetings (removeMeeting, getMeeting)
- Input validation via checkTimes (month, day, start/end hour ranges)

2.Fixes Applied to Test Code

Running the tests for the first time revealed NullPointerExceptions in several test cases. The root cause was the four-argument Meeting constructor Meeting(month, day, start, end), which leaves both the description field and the room field as null. Two code paths in the system then crash on these nulls:

- Calendar.addMeeting() calls toCheck.getDescription().equals("Day does not exist") crashes when description is null.
- Meeting.toString() calls room.getID() crashes when room is null.

Fix applied across CalendarTest, PersonTest, and RoomTest: a private helper method makeMeeting(month, day, start, end) was introduced in each class. It calls the full seven-argument constructor, supplying an empty attendee list, a named Room object, and the description string "test meeting". All test cases that previously used the four-arg constructor now call makeMeeting() instead.

3.New Bugs Discovered During Test Execution

Beyond the five pre-documented faults, two additional behavioural bugs were discovered when the tests actually ran:

Bug ID	Title	Root Cause & Impact	Discovered By
--------	-------	---------------------	---------------

GROUP D (EVE)

NB-01	Calendar. Feb 29 not blocked	The addMeeting conflict check intentionally skips entries with description 'Day does not exist', so any new meeting bypasses the Feb 29 placeholder and is accepted.	TC-CAL-07
NB-02	Calendar. April 31 not blocked	Same root cause as NB-01. isBusy skips the 'Day does not exist' entry on April 31 and returns false, giving a misleading free-slot result.	TC-CAL-18
NB-03	Meeting four-arg constructor	Meeting(month,day,start,end) leaves description=null and room=null. Calling addMeeting() with such a Meeting causes a NullPointerException inside Calendar.addMeeting() and Meeting.toString(). Fix: always supply description and room.	All NPE errors in run 1

4. Known Faults (from Lab Slides)

The five faults documented in the lab slides are each targeted by at least one dedicated test case. All five are confirmed by the current test run.

Fault ID	Location	Description	Exposed By
Fault 1	getMeeting / removeMeeting	No bounds/date validation, any index accepted without error	TC-CAL-20, TC-PER-17, TC-ROM-20, TC-ROM-21
Fault 2	Calendar constructor	Loop uses $i \leq 13$, creating 14 months (indices 0–13). Month 13 should not exist.	TC-CAL-21
Fault 3	Calendar constructor	Nov 30 placeholder: Meeting(11,31,...) stored at day index 30, day field mismatch.	TC-CAL-25
Fault 4	checkTimes	$mMonth \geq 12$ rejects December (month 12). Condition should be $mMonth > 12$.	TC-CAL-04, TC-PER-07, TC-ROM-08
Fault 5	checkTimes	$mStart \geq mEnd$ rejects same-hour meetings. Should be $mStart > mEnd$.	TC-CAL-11

5. CalendarTest: 25 Test Cases

Class under test: edu.sc.bse3211.meetingplanner.Calendar

File: CalendarTest.java

Test ID	Feature	Description	Input	Expected	Status
TC-CAL-01	addMeeting	Add full-day holiday	Meeting(2,16,'Janan Luwum')	isBusy(2,16,0,23)=true	PASS

GROUP D (EVE)

TC-CAL-02	addMeeting	Add valid timed meeting	makeMeeting(3,15,9,11)	isBusy(3,15,9,11)=true	PASS
TC-CAL-03	addMeeting	Two non-overlapping meetings same day	makeMeeting(1,10,8,10) + makeMeeting(1,10,11,13)	Both busy; gap 10-11 free	PASS
TC-CAL-04	addMeeting	December 25 (Fault 4: >= 12 rejects Dec)	makeMeeting(12,25,10,12)	Should succeed but FAILS due to bug	FAIL (Bug)
TC-CAL-05	addMeeting	Added meeting is retrievable	makeMeeting(5,20,14,16) → getMeeting(5,20,0)	start=14, end=16	PASS
TC-CAL-06	addMeeting	Overlapping meetings → exception	makeMeeting(4,10,9,12) then (4,10,10,14)	TimeConflictException thrown	PASS
TC-CAL-07	addMeeting	Feb 29 — blocking doesn't work	makeMeeting(2,29,9,10)	BUG: meeting accepted (placeholder skipped)	BUG DOCUMENTED
TC-CAL-08	addMeeting	Day = 35 (out of range)	makeMeeting(2,35,9,10)	TimeConflictException thrown	PASS
TC-CAL-09	addMeeting	Day = 0 (below minimum)	makeMeeting(3,0,9,10)	TimeConflictException thrown	PASS
TC-CAL-10	addMeeting	Month = 0 (below minimum)	makeMeeting(0,5,9,10)	TimeConflictException thrown	PASS
TC-CAL-11	addMeeting	Start == End (Fault 5: >= instead of >)	makeMeeting(3,10,9,9)	Should be allowed — FAILS due to bug	FAIL (Bug)
TC-CAL-12	addMeeting	Start after end	makeMeeting(6,15,14,10)	TimeConflictException thrown	PASS
TC-CAL-13	isBusy	Empty slot returns false	isBusy(1,15,9,11) on fresh calendar	false	PASS
TC-CAL-14	isBusy	True after addMeeting	addMeeting then isBusy same slot	true	PASS
TC-CAL-15	isBusy	Invalid month = 13	isBusy(13,1,9,11)	TimeConflictException thrown	PASS
TC-CAL-16	isBusy	End hour = 25 (invalid)	isBusy(3,10,0,25)	TimeConflictException thrown	PASS
TC-CAL-17	isBusy	Partial overlap detected	Add 9-14; query 12-16	true	PASS
TC-CAL-18	isBusy	April 31, blocking doesn't work	isBusy(4,31,9,11)	BUG: returns false instead of throwing	BUG DOCUMENTED
TC-CAL-19	removeMeeting	Slot freed after remove	addMeeting → removeMeeting → isBusy	false after removal	PASS
TC-CAL-20	getMeeting	Invalid index, Fault 1	getMeeting(1,15,99)	IndexOutOfBoundsException	PASS
TC-CAL-21	getMeeting	Month 13 accessible, Fault 2	getMeeting(13,1,0)	Month 13 exists in array meaning bug confirmed	FAIL (Bug)
TC-CAL-22	clearSchedule	All meetings removed for day	addMeeting x2 → clear → isBusy	false for both slots	PASS

GROUP D (EVE)

TC-CAL-23	printAgenda	Month agenda, non-null string	printAgenda(3)	Non-null, contains '3'	PASS
TC-CAL-24	printAgenda	Day agenda contains date	addMeeting(5,10,9,11) → printAgenda(5,10)	Contains '5/10'	PASS
TC-CAL-25	Calendar init	Nov 30 day-field mismatch, Fault 3	getMeeting(11,30,0).getDay()	Returns 31 implying mismatch confirmed	FAIL (Bug)

6. MeetingTest:14 Test Cases

Class under test: edu.sc.bse3211.meetingplanner.Meeting

File: MeetingTest.java

Test ID	Feature	Description	Input	Expected	Status
TC-MTG-01	Constructor	Default constructor having zero fields	new Meeting()	month=0, day=0	PASS
TC-MTG-02	Constructor	Two-arg with full day block	new Meeting(4,15)	start=0, end=23	PASS
TC-MTG-03	Constructor	Three-arg with description	new Meeting(1,1,'New Year')	description='New Year'	PASS
TC-MTG-04	Constructor	Four-arg with time range	new Meeting(6,10,9,17)	start=9, end=17	PASS
TC-MTG-05	Constructor	Full constructor with all fields	new Meeting(3,21,10,12,list,room,'Sprint')	All fields correctly set	PASS
TC-MTG-06	Setter/Getter	setMonth / getMonth	m.setMonth(7)	getMonth()=7	PASS
TC-MTG-07	Setter/Getter	setDay / getDay	m.setDay(22)	getDay()=22	PASS
TC-MTG-08	Setter/Getter	setStartTime / getStartTime	m.setStartTime(8)	getStartTime()=8	PASS
TC-MTG-09	Setter/Getter	setEndTime / getEndTime	m.setEndTime(17)	getEndTime()=17	PASS
TC-MTG-10	Setter/Getter	setDescription / getDescription	m.setDescription('Board')	getDescription()='Board'	PASS
TC-MTG-11	Attendees	addAttendee adds person	m.addAttendee(person)	contains(person)=true	PASS
TC-MTG-12	Attendees	removeAttendee removes person	m.removeAttendee(person)	contains(person)=false	PASS

GROUP D (EVE)

TC-MTG-13	Attendees	Three attendees with correct count	Add 3 attendees	getAttendees().size()==3	PASS
TC-MTG-14	toString	toString includes room, desc, name	Meeting with room+desc+attendee	Contains room ID, description, name	PASS

7. PersonTest: 20 Test Cases

Class under test: edu.sc.bse3211.meetingplanner.Person

File: PersonTest.java

Test ID	Feature	Description	Input	Expected	Status
TC-PER-01	Constructor	Default constructor with empty name	new Person()	getName()=="	PASS
TC-PER-02	Constructor	Named constructor stores name	new Person('Namugga Martha')	getName()='Namugga Martha'	PASS
TC-PER-03	addMeeting	Valid timed meeting - no exception	makeMeeting(3,10,9,11)	No exception	PASS
TC-PER-04	addMeeting	Full-day block marks person busy	Meeting(8,5,'Annual Leave')	isBusy(8,5,0,23)=true	PASS
TC-PER-05	addMeeting	Two non-overlapping same day	makeMeeting(5,20,8,10) + (5,20,14,16)	Both slots busy	PASS
TC-PER-06	addMeeting	Added meeting retrievable	makeMeeting(4,15,10,12) → getMeeting	start=10, end=12	PASS
TC-PER-07	addMeeting	December - Fault 4 exposed	makeMeeting(12,15,9,11)	Should succeed — FAILS due to bug	FAIL (Bug)
TC-PER-08	addMeeting	Overlap message contains person name	Add overlap → catch exception	Message contains 'Namugga Martha'	PASS
TC-PER-09	addMeeting	Day = 0 throws	makeMeeting(3,0,9,11)	TimeConflictException	PASS
TC-PER-10	addMeeting	Month = 0 throws	makeMeeting(0,10,9,11)	TimeConflictException	PASS
TC-PER-11	addMeeting	Start after end throws	makeMeeting(5,10,15,9)	TimeConflictException	PASS
TC-PER-12	isBusy	Fresh person - not busy	isBusy(1,15,9,11) on new Person	false	PASS
TC-PER-13	isBusy	True after addMeeting	addMeeting → isBusy	true	PASS

GROUP D (EVE)

TC-PER-14	isBusy	End hour > 23 throws	isBusy(3,10,0,25)	TimeConflictException	PASS
TC-PER-15	isBusy	Partial overlap detected	Add 9-14; query 12-16	true	PASS
TC-PER-16	removeMeeting	Slot freed after remove	addMeeting → removeMeeting → isBusy	false	PASS
TC-PER-17	removeMeeting	Invalid index - Fault 1	removeMeeting(1,10,99)	IndexOutOfBoundsException	PASS
TC-PER-18	printAgenda	Month agenda non-null	printAgenda(3)	Non-null, contains '3'	PASS
TC-PER-19	printAgenda	Day agenda contains date	addMeeting(5,18,9,11) → printAgenda(5,18)	Contains '5/18'	PASS
TC-PER-20	printAgenda	Empty day contains date	printAgenda(1,10)	Contains '1/10'	PASS

8. RoomTest: 25 Test Cases

Class under test: edu.sc.bse3211.meetingplanner.Room

File: RoomTest.java

Test ID	Feature	Description	Input	Expected	Status
TC-ROM-01	Constructor	Default constructor with an empty ID	new Room()	getID()=""	PASS
TC-ROM-02	Constructor	Named constructor stores ID	new Room('LLT6A')	getID()='LLT6A'	PASS
TC-ROM-03	addMeeting	Valid timed meeting, no exception	makeMeeting(3,10,9,11)	No exception	PASS
TC-ROM-04	addMeeting	Full-day block marks room busy	Meeting(6,1,'Maintenance')	isBusy(6,1,0,23)=true	PASS
TC-ROM-05	addMeeting	Two non-overlapping same day	makeMeeting(4,12,8,10)+(4,12,13,15)	Both slots busy	PASS
TC-ROM-06	addMeeting	Added meeting retrievable	makeMeeting(7,22,14,16) → getMeeting	start=14, end=16	PASS

GROUP D (EVE)

TC-ROM-07	addMeeting	Different days so no conflict	(5,10,9,11) + (5,11,9,11)	Both days busy independently	PASS
TC-ROM-08	addMeeting	December and Fault 4 is exposed	makeMeeting(12,10,9,11)	Should succeed but FAILS due to bug	FAIL (Bug)
TC-ROM-09	addMeeting	Overlap message contains room ID	Add overlap → catch exception	Message contains 'LLT6A'	PASS
TC-ROM-10	addMeeting	Day = 0 throws	makeMeeting(3,0,9,11)	TimeConflictException	PASS
TC-ROM-11	addMeeting	Day = 32 throws	makeMeeting(3,32,9,11)	TimeConflictException	PASS
TC-ROM-12	addMeeting	Month = 0 throws	makeMeeting(0,10,9,11)	TimeConflictException	PASS
TC-ROM-13	addMeeting	Start after end throws	makeMeeting(5,10,15,9)	TimeConflictException	PASS
TC-ROM-14	isBusy	Fresh room, not busy	isBusy(1,15,9,11) on new Room	false	PASS
TC-ROM-15	isBusy	True after addMeeting	addMeeting → isBusy	true	PASS
TC-ROM-16	isBusy	Free outside booked slot	Add 10-12; query 13-15	false	PASS
TC-ROM-17	isBusy	End hour > 23 throws	isBusy(4,10,0,24)	TimeConflictException	PASS
TC-ROM-18	isBusy	Partial overlap detected	Add 9-13; query 11-15	true	PASS
TC-ROM-19	removeMeeting	Slot freed after remove	addMeeting → removeMeeting → isBusy	false	PASS
TC-ROM-20	removeMeeting	Invalid index - Fault 1	removeMeeting(1,10,99)	IndexOutOfBoundsException	PASS
TC-ROM-21	getMeeting	Invalid index - Fault 1	getMeeting(1,10,99)	IndexOutOfBoundsException	PASS
TC-ROM-22	printAgenda	Month agenda non-null	printAgenda(5)	Non-null, contains '5'	PASS
TC-ROM-23	printAgenda	Day agenda contains date	addMeeting(6,14,9,11) → printAgenda(6,14)	Contains '6/14'	PASS
TC-ROM-24	printAgenda	Empty day contains date	printAgenda(2,20)	Contains '2/20'	PASS
TC-ROM-25	Independent	Two rooms have independent calendars	Book LLT6A; check LLT6B	LLT6B isBusy=false	PASS

9. OrganizationTest: 22 Test Cases

Class under test: edu.sc.bse3211.meetingplanner.Organization

File: OrganizationTest.java

Organization initialises 5 pre-loaded employees (Namugga Martha, Shema Collins, Acan Brenda, Kazibwe Julius, Kukunda Lynn) and 5 rooms (LLT6A, LLT6B, LLT3A, LLT2C, LAB2). Tests verify list contents, successful lookup, error handling for unknown/invalid inputs, case-sensitivity, object identity on repeated calls, and calendar independence between entities.

Test ID	Feature	Description	Input	Expected	Status
TC-ORG-01	getEmployees	Returns non-null list	org.getEmployees()	Not null	PASS
TC-ORG-02	getEmployees	Returns exactly 5 employees	org.getEmployees().size()	5	PASS
TC-ORG-03	getEmployees	Contains all 5 expected names	Check all names in list	All 5 names present	PASS
TC-ORG-04	getRooms	Returns non-null list	org.getRooms()	Not null	PASS
TC-ORG-05	getRooms	Returns exactly 5 rooms	org.getRooms().size()	5	PASS
TC-ORG-06	getRooms	Contains all 5 expected IDs	Check all IDs in list	LLT6A,LLT6B,LLT3A,LLT2C,LAB2 present	PASS
TC-ORG-07	getEmployee	Known name returns correct Person	getEmployee('Namugga Martha')	Person with name='Namugga Martha'	PASS
TC-ORG-08	getEmployee	All 5 employees found by name	getEmployee() for each name	All return non-null with correct name	PASS
TC-ORG-09	getEmployee	Unknown name throws Exception	getEmployee('Nobody Here')	Exception thrown	PASS
TC-ORG-10	getEmployee	Empty string throws Exception	getEmployee("")	Exception thrown	PASS
TC-ORG-11	getEmployee	Null name throws or handles gracefully	getEmployee(null)	Any exception thrown	PASS
TC-ORG-12	getEmployee	Wrong case throws Exception	getEmployee('namugga martha')	Exception thrown (case-sensitive)	PASS
TC-ORG-13	getEmployee	Repeated call returns same instance	getEmployee() twice for same name	assertSame — same object	PASS
TC-ORG-14	getRoom	Known ID returns correct Room	getRoom('LLT6A')	Room with ID='LLT6A'	PASS
TC-ORG-15	getRoom	All 5 rooms found by ID	getRoom() for each ID	All return non-null with correct ID	PASS
TC-ORG-16	getRoom	Unknown ID throws Exception	getRoom('ROOM999')	Exception thrown	PASS
TC-ORG-17	getRoom	Empty string throws Exception	getRoom("")	Exception thrown	PASS

GROUP D (EVE)

TC-ORG-18	getRoom	Null ID throws or handles gracefully	getRoom(null)	Any exception thrown	PASS
TC-ORG-19	getRoom	Wrong case throws Exception	getRoom('llt6a')	Exception thrown (case-sensitive)	PASS
TC-ORG-20	getRoom	Repeated call returns same instance	getRoom() twice for same ID	assertSame — same object	PASS
TC-ORG-21	Independence	Employees have independent calendars	Book Martha; check Collins	Collins isBusy=false	PASS
TC-ORG-22	Independence	Rooms have independent calendars	Book LLT6A; check LLT6B	LLT6B isBusy=false	PASS

10. Test Summary

Total Tests	Passed	Failed (Bugs)	Bugs Documented	Faults Covered
106	101	5	3	5 / 5

```
[INFO]
[ERROR] Tests run: 106, Failures: 5, Errors: 0, Skipped: 0
[INFO]
```

Figure 1: Screenshot from the terminal `mvn test` showing the test summary

Tests marked FAIL (Bug) are intentionally failing, they expose real defects in the source code and will pass once those defects are corrected. Tests marked BUG DOCUMENTED reveal additional bugs discovered during test execution; they are written to assert the current (broken) behavior so the run does not crash, while clearly labelling the problem for the developer.